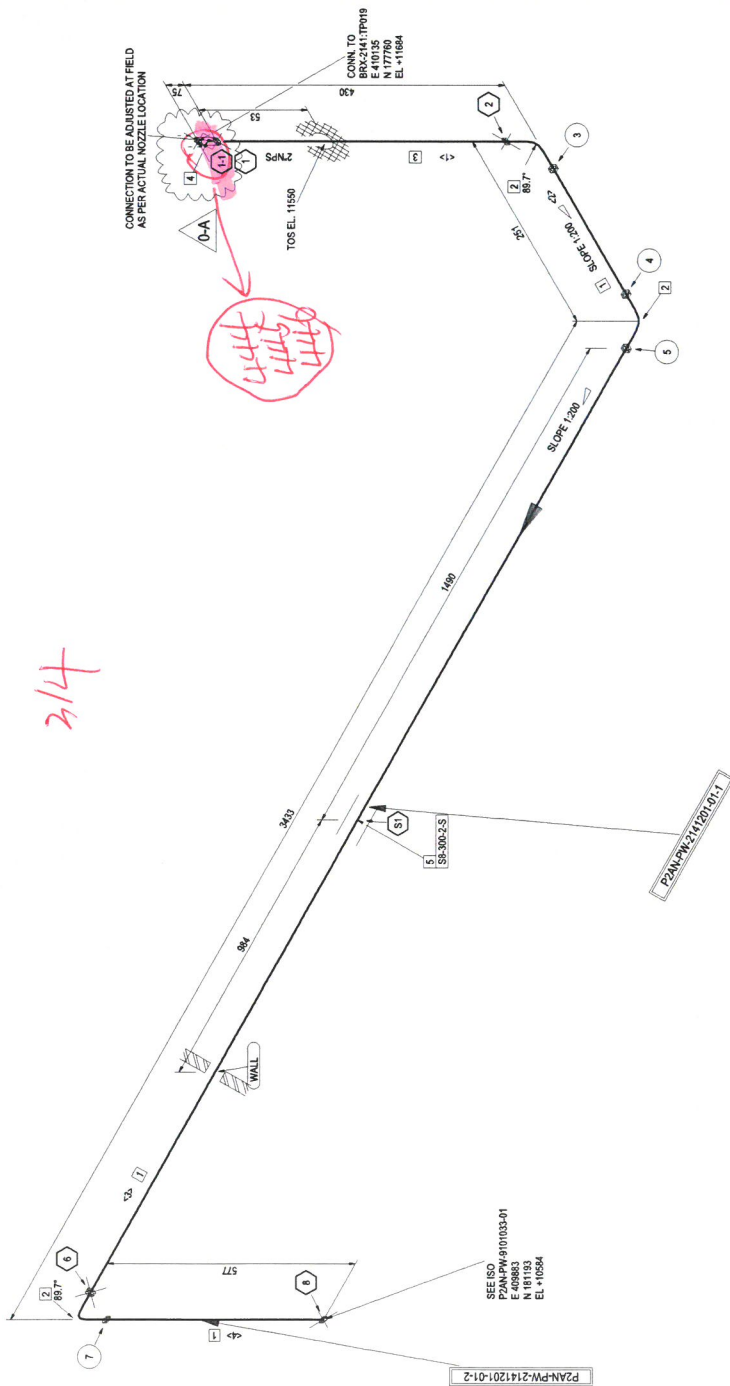


XI-JHC-910-PW01-007

FABRICATION MATERIALS						
NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SAFETY CODE	ITEM
1	PPE EVA ASTM A132-TP304L SCH15 BE RT-100%	2"	3.8 M	S99	P	PA62-00000223
2	90 DEG ELBOW (B) BW WELDED ASTM A486-TP304L SCH15	2"	3	S99	E	PR59-00035417
EJECTION MATERIALS						
NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SAFETY CODE	ITEM
3	PPE EVA ASTM A132-TP304L SCH15 BE RT-100%	2"	6.4 M	S99	P	PA62-00000223
4	PPE NIPPLE BEE SMLS ASTM A132-TP304L SCH15 TRANS (J) TRANSITION	2"	1	S99	NBBD	PR59-001126117
5	ECE TO ASME BPE	2"	1			S9
CUT PIPE LENGTH						
	NO	LENGTH	WPD	REMARKS	WPD	REMARKS
1	1	354	(IN)		(IN)	
2	2	354	2		2	
3	2	3302	2		2	
4	2	3302	2		2	



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[illegible]

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PIPING MANAGER Bea Chang Won
2026-01-26

LOTTE BIOLOGICS

8	CWA NO.	P2AN
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WG NO.	P2AN-PW-2141201-01
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PREPARED	CHECKED	APPROVED
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DESCRIPTION

REV.	DATE
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[illegible]

[illegible]

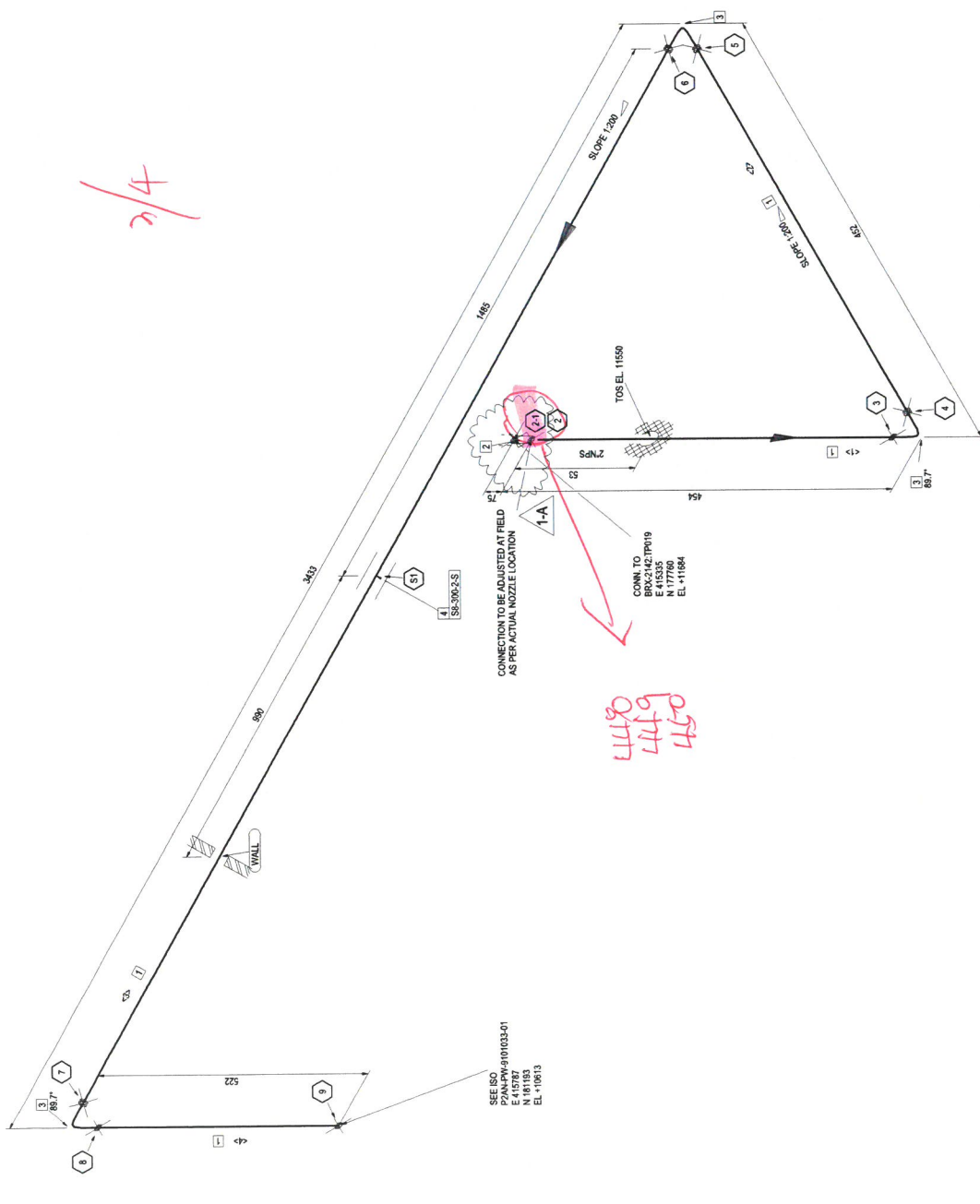
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988



PIPE NAME: 2-PW-2141201-S99-1SN_P2AN-01

EXPERIMENTAL MATERIALS

[illegible]

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[illegible]

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2026-01-26



LOTTE BIOLOGICS

 LOTTE F&C

PROJECT PROVIDENCE

AG	PLANT
...	...

NO.	P2AN-PW-2142201-01
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1 OF 1	REV.	1-A
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SHIT NO.

1

1

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1-21422

P2AN-

WAG NO.

APPROVED

CHECKED

PREPARED

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1001

0530

[illegible]

DATE _____

9

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[illegible][illegible]

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[illegible][illegible][illegible]

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[illegible]

1

[illegible]

F3097



PIPE MODIFIED AT 11:06 26 JUL 2025, DRAWING CREATED AT 11:06 20 JUL 2025
PIPE NAME: 2-PW-2142201-S99-1SN_P2AN-01

FABRICATION MATERIALS

COMPONENT DESCRIPTION				SIZE	QTY	MAT CLASS	SHORT CODE	ITEM
1	PIPE	PIPE ASTM A153 TP30400 SCH 80	RT-100%	2"	3.54	598	P	PA200000223
2	90 DEG ELBOW	90 DEG ELBOW 1/2" NPT WELDED ASTM A182-30400 SCH 80		2"	1	598	E	P9590005817

SECTION MATERIALS				SIZE	QTY	MAT CLASS	SHORT CODE	ITEM
COMPONENT DESCRIPTION								
1	PIPE	PIPE ASTM A153 TP30400 SCH 80	RT-100%	2"	0.54	598	P	PA200000223
2	FLANGE	FLANGE 1/2" NPT WELDED ASTM A182-30400 SCH 80		2"	1	598	NESD	P9590005817
3	SA-106-A S			2"	1			58

CUT PIPE LENGTH				PRICE	LENGTH	WFO	REMARKS	WFO	REMARKS
					(MM)	(IN)		(IN)	
1	15'	456	2	<2	250	2			
2	3281	2	2	<4	359	2			

1988

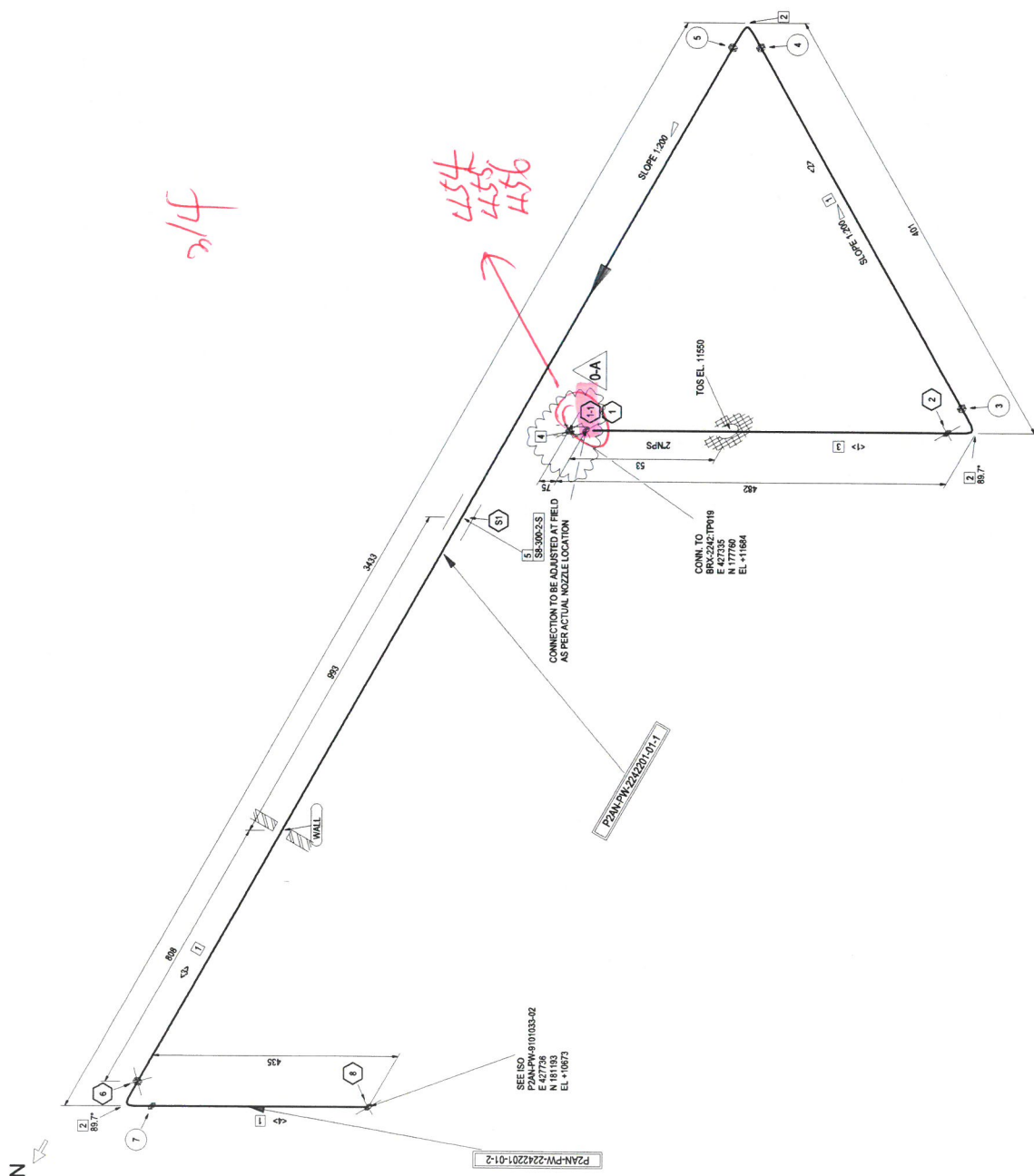


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2026-01-26


LOTTE BIOLOGICS

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NO.	AG	PLANT	N	AREA	POB	OWA NO.	P2AN
NO.	K1-DSL-PR-PID-2242			PROL NO.	16480	FLUID	PW
NO.	-			LINE NO.	2-PW-224201-S99-SIN		
NO.	P2AN-PW-224201-01					SHT NO.	1 OF 1 REV 0-A



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[illegible]

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BILL OF MATERIALS

FABRICATION MATERIALS

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
2	90 DEG ELBOW (B) BW WELDED ASTM A13.3 TYPICAL SCH30S	2"	3	S99	E	P2AN-20000847

ERECTION MATERIALS

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
3	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
4	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
5	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
6	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
7	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
8	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
9	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
10	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
11	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
12	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
13	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
14	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
15	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
16	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
17	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
18	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
19	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
20	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
21	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
22	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
23	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
24	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
25	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
26	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
27	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
28	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
29	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
30	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
31	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
32	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
33	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
34	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
35	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
36	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
37	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
38	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
39	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
40	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
41	BES TO ASME BPE	2"	1			58

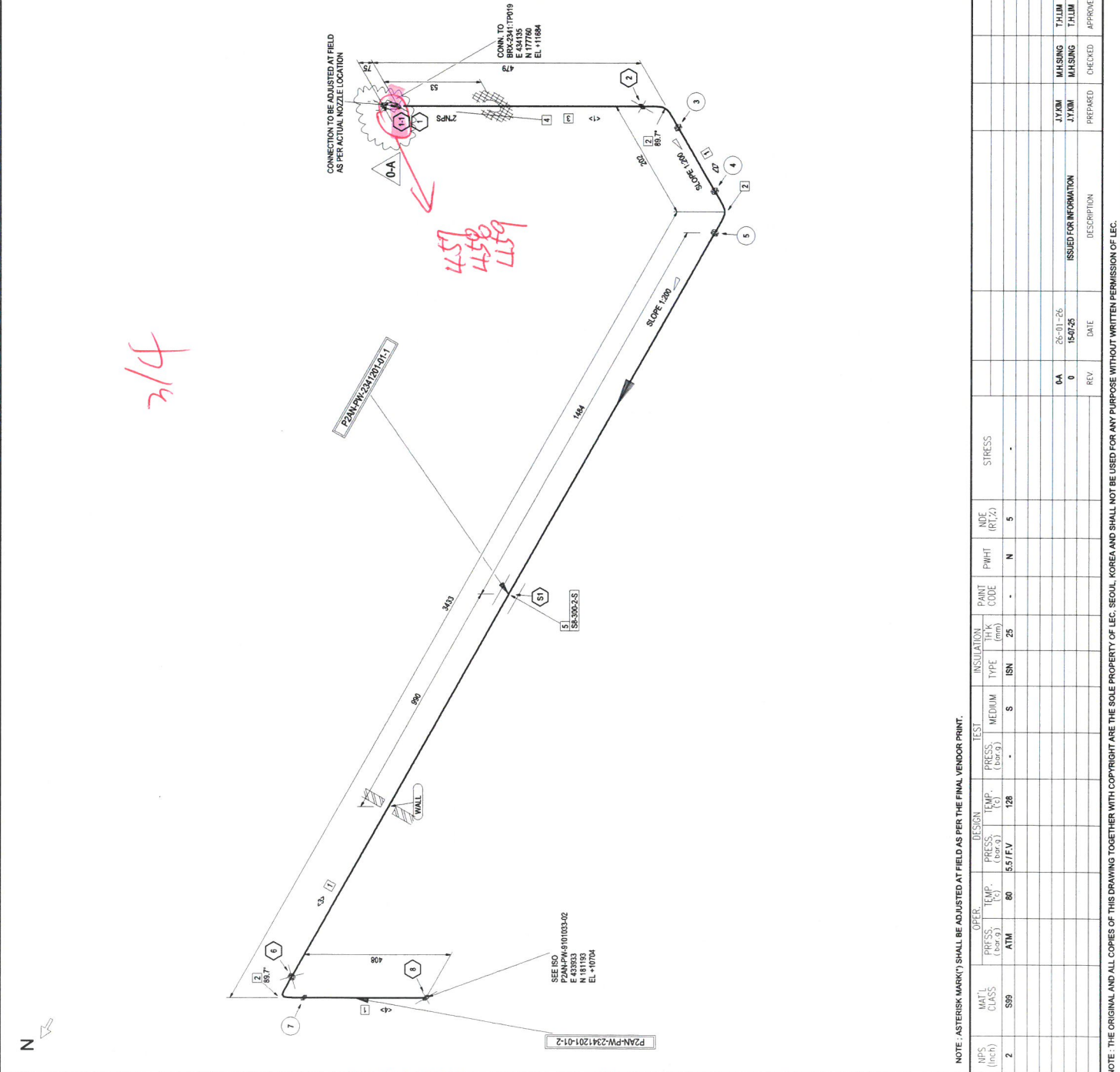
NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
42	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
43	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
44	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
45	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
46	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
47	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
48	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
49	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
50	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
51	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
52	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
53	BES TO ASME BPE	2"	1			58

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
54	PPE EPPA ASTM A13.3 TYPICAL SCH30S BE RT-100%	2"	1	S99	P	P2AN-20000723
55	PPE MPPE EBE SMALL ASTM A13.3 TYPICAL SCH30S TRANSITION P	2"	1	S99	MED	P2AN-20011817
56	BES TO ASME BPE	2"	1			58



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2026-01-26

PROJECT		LOTTE BIOLOGICS	
PROJECT PROVIDENCE		LOTTE E&C	
PART	AG	PLANT	P2AN
P&ID NO.	K1-OSL-PR-PID-2341	AREA	PW
CAD NO.	-	PROJ. NO.	16400
		LINE NO.	2-PW-2341201-599-1SN
DWG. NO.	P2AN-PW-2341201-01	SHT. NO.	1 OF 1
		REV	1

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FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
	PIPE EPW ASTM A312 TP304/304L SCH10S BE RT=100%	2"	3.9 M	S99	P	PA02-000001723
	90 DEG. ELBOW (LR) BW WELDED ASTM A403 WP304/304L SCH10S	2"	3	S99	E	P899-0001038417

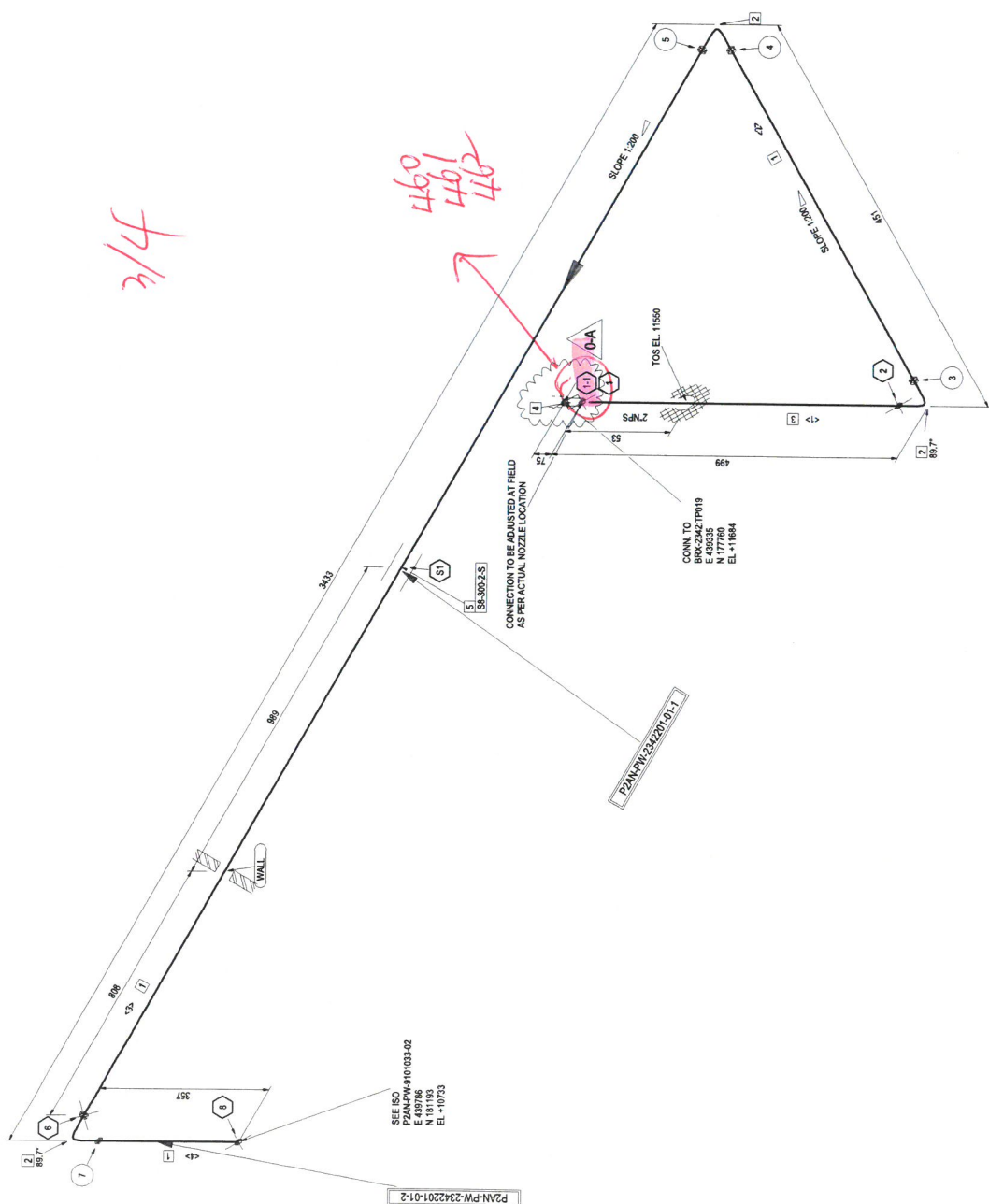
SECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
	PPE EDW ASTM A312 TP304/304L SCH10S BE RT-100%	2"	0.5 M	S99	P	PA02-000001723
	PPE NPPLE BBE ASTM A312 TP304/304L SCH10S 7MM(0.07" TRANSITION P	2"	1	S99	NBS0	P699-000139517

NOTE TO FABRICATOR

ITEM NO.	DESCRIPTION	UNIT	QUANTITY	PRICE	TOTAL
58-300-2.5	OUT PIPE LENGTH	1	1	58	58

Q	(MM)	(N)	NO	(MM)	(N)	REMARKS
13	424	2	<2	300	2	
20	3281	2	<4	281	2	



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2026-01-26

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LOTTE BIOLOGICS

PROJECT PROVIDENCE

UNIT NO.	AG	PLAN#	N	AREA	POB	CWA NO.	P2AN
NO.	K1-OSL-PP-PD-2342			PROJ NO.	16480		PW
	-			LINE NO.	2-PW-234201-S90-LSN		
NO.						SHT NO.	REV
						1 OF 1	0-A

BILL OF MATERIALS

FABRICATION MATERIALS

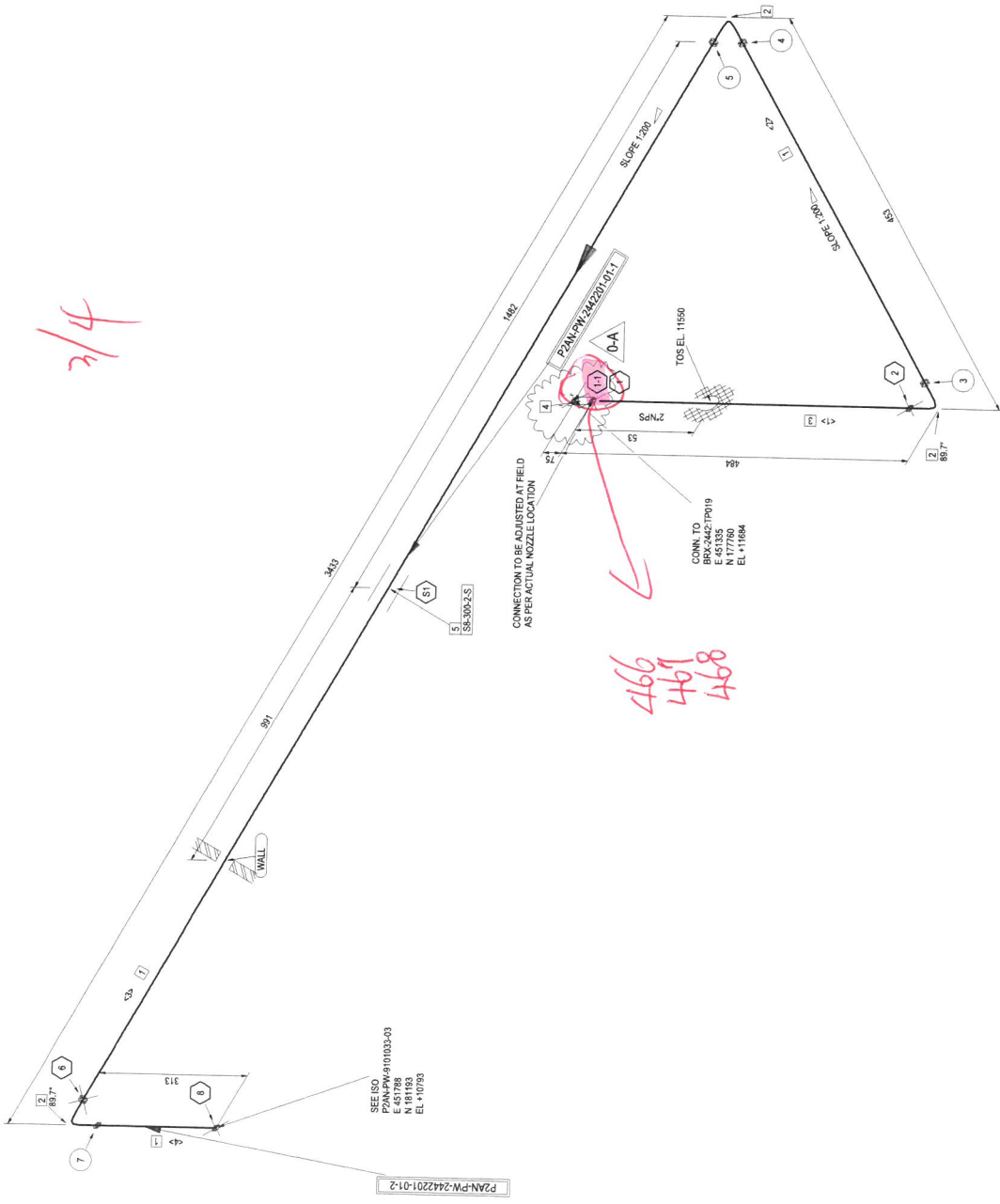
NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PPE ERM ASTM A312 TYPICAL SCH40S BE PT-100%	2"	33M	S99	P	PA20-00000723
2	90 DEG ELBOW (B) BAWELDED ASTM A452 WP304/304L SCH40S	2"	3	S99	E	PP89-00038417

ERECTION MATERIALS

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
3	PPE ERM ASTM A312 TYPICAL SCH40S BE PT-100%	2"	0.5M	S99	P	PA20-00000723
4	PPE NPPV E BBE SMLS ASTM A312 TP304/304L SCH40S TRANSITION P	2"	1	S99	NBDO	PP89-00119617
5	ICE TO ASME BPE	2"	1			S9

CUT PIPE LENGTH

PIECE NO.	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO.	LENGTH (MM)	NPD (IN)	REMARKS
<1>	408	2		<2>	301	2	
<3>	281	2		<4>	237	2	



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2026-01-26



PROJECT

PROJECT PROVIDENCE

PART	AG	PLANT	N	AREA	PDB	CDW NO.	P2AN
PA20-00000723	K1-DL-PR-PID-2442			16480		2442201-01-01	PW



DATE	REV	DESCRIPTION	PREPARED	CHECKED	APPROVED	DATE	REV	DESCRIPTION
26-01-26	0	ISSUED FOR INFORMATION	JYHM	JYHM	JYHM	15-07-25	0	ISSUED FOR INFORMATION

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